Software Documentation

1. Software Overview

The software controls the full functionality of the WiFi-enabled toy car. It enables communication between the mobile app and the ESP32, allowing the user to control movement, LED indicators, and the reverse assist system.

- The **ESP32 (MicroPython)** processes incoming commands and controls motors, LEDs, ultrasonic sensor, and buzzer.
- The **mobile app (MIT App Inventor)** acts as a remote control, sending commands over WiFi.

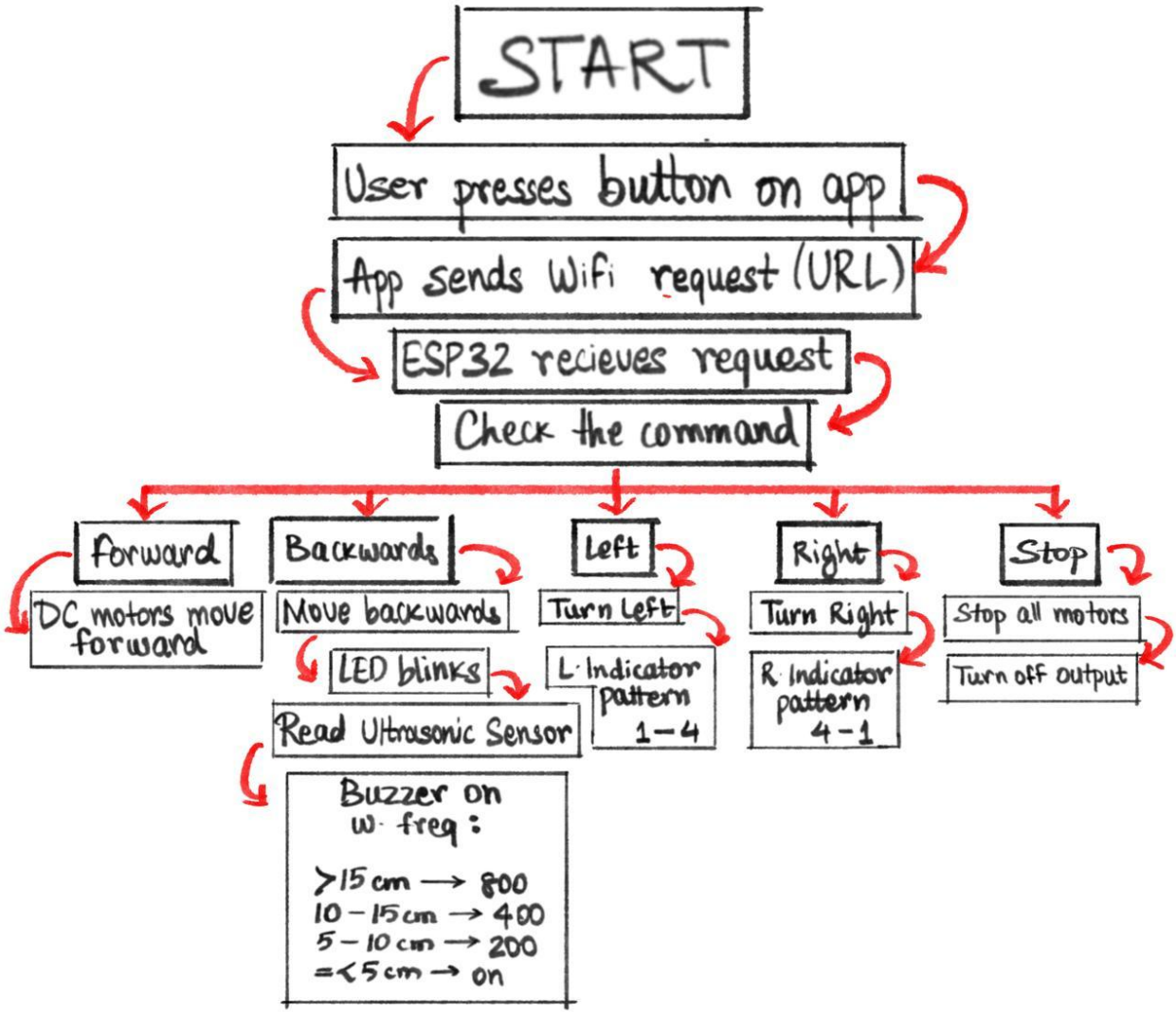
Platforms Used:

- MicroPython (ESP32 programming in Thonny)
- MIT App Inventor (mobile app interface)
- Cirkit Designer (wiring and testing)

2. System Logic

1. User presses a button on the mobile app
2. App sends a command via WiFi (HTTP request)
3. ESP32 receives the command
4. Based on the command:
    - Motors move (forward/back/left/right/stop)
    - LEDs activate (turn indicators or reverse blinking)
5. If reversing:
    - Ultrasonic sensor measures distance continuously
    - Buzzer frequency changes based on distance
6. System updates outputs in real time

3. Flowchart


START
User presses button on app
App sends Wifi request (URL)
ESP32 recieves request
Check the command
Forward
DC motors move forward
Backwards
Move backwards
LED blinks
Read Ultrasonic Sensor
Buzzer on w. freq :
>15 cm → 800
10 – 15 cm → 400
5 – 10 cm → 200
=<5 cm → on
Left
Turn Left
L. Indicator pattern 1 – 4
Right
Turn Right
R. Indicator pattern 4 – 1
Stop
Stop all motors
Turn off output

4. App Flow (If MIT App Inventor)

Screen Flow:

- Single screen interface
- Contains 5 main buttons: Forward, Backward, Left, Right, Stop

Button Actions:

Button	**Action**
Forward	Sends /F → Car moves forward
Backward	Sends /B → Car reverses + LEDs blink + buzzer activates
Left	Sends /L → Car turns left + left LEDs blink
Right	Sends /R → Car turns right + right LEDs blink
Stop	Sends /S → Car stops

Data Flow:

1. Button click → sets URL (e.g., http://192.168.4.1/F)
2. App sends request using Web component (Web1.Get)
3. ESP32 receives request via WiFi server
4. ESP32 executes corresponding function
5. Physical components respond (motors, LEDs, buzzer)

5.Final Code Used:

https://github.com/naishark06/ODT_Group-Work_Repo/blob/main/Final_Final_Car_Code.py